

Rand Protocol

Executive Summary for Investors

Building the Future of Private, Scalable Blockchain Infrastructure

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Confidential Investment Materials

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1 Executive Summary

Rand is a privacy layer for Solana — not a new blockchain.

Solana settles more stablecoin volume than any other high-performance chain, and native USDT and USDC are issued directly on it. Everything is also completely public: every balance, every counterparty, every amount. That rules Solana out for payroll, treasury operations, supplier payments, and trading — any activity a business cannot publish.

Rand fixes exactly that, and nothing else:

1. **Private transfers of the stablecoins people already hold.** Deposit native USDT, transact confidentially, withdraw. No bridge, no wrapped assets, no new chain to bootstrap.
2. **Deployed as Solana programs.** We inherit Solana's speed, finality, liquidity, and validator set instead of rebuilding them. No validators to recruit, no security budget to fund, no chicken-and-egg liquidity problem.
3. **Post-quantum confidentiality.** Encrypted transfer data on a public ledger is permanently exposed to future quantum decryption. Rand encrypts with the NIST post-quantum standards from day one.

Why this is the right structure: a privacy Layer 1 must attract its own liquidity, its own validators, and its own stablecoin issuers before it is useful. Rand starts with all three, because they are already on Solana.

2 The Problem

2.1 Problem 1: Solana Has No Privacy

Imagine if every bank transfer your company made was permanently visible to competitors, suppliers, and employees. That is Solana today. Balances, counterparties, and amounts are public by default.

Existing options all fail for a business holding USDT on Solana:

- **Zcash, Monero:** strong privacy, but separate networks. Your USDT is not there, and getting it there means bridging.
- **Token-2022 confidential transfers:** hides amounts but not who pays whom — and cannot be enabled on the existing USDT and USDC mints, because the feature must be set when a token is created.
- **Privacy rollups:** own gas token and own liquidity, so assets must be bridged and liquidity fragmented.

2.2 Problem 2: Privacy Solutions Force You Off-Chain

Every option above asks a business to move assets somewhere else. Bridges are the single largest category of loss in crypto history — Ronin \$624M, Poly \$611M, BNB Bridge \$570M, Wormhole \$325M. Asking a treasurer to bridge their stablecoins to obtain privacy is asking them to accept a worse risk than the one they are solving.

2.3 Problem 3: Tomorrow's Quantum Computers Read Today's Ledger

Encrypted data published to a blockchain stays there permanently. An adversary can record it now and decrypt it once quantum computers mature — "harvest now, decrypt later." Privacy systems using conventional encryption are therefore not private in the long run; they are merely delayed.

3 Rand's Solution

Rand adds three Solana programs and one off-chain role. That is the whole system.

3.1 The Shielded Pool

What it does: holds deposited USDT, USDC, and SOL, and tracks who owns what without revealing it.

Deposits are visible. Once inside, transfers are not: sender, recipient, and amount are hidden behind a zero-knowledge proof. Withdrawals are visible again. All assets are the real, native tokens throughout — never a wrapped derivative.

3.2 The Relayers (Earn SHRUGG)

What they do: submit your private transaction and pay the Solana network fee on your behalf.

This role exists for a precise reason. Solana charges fees in SOL, and every transaction names who paid. If you pay your own fee, you have signed your name to your "private" transfer and defeated the entire system. So somebody else must submit it — and be paid without creating a second trail. That payment is SHRUGG, deducted inside the zero-knowledge proof.

- **Requirement:** an account with SOL. No stake, no permission, no special hardware.
- **Payment:** SHRUGG, from every transfer they submit
- **Competition:** relayers are interchangeable and cannot alter what they submit — only submit it or decline

Why SHRUGG has durable demand: it is not a reward paid out of inflation. Every confidential transfer must buy it to pay a relayer, so demand scales with usage rather than with emissions.

3.3 Solana Itself

What it provides: ordering, finality, security, and liquidity — for free, from our perspective.

Solana validators run unmodified Agave or Firedancer and need not know Rand exists. We are a program on their chain, not a competitor for their stake.

- **Speed:** Solana's, not a number we have to prove
- **Security:** secured by the full value staked on Solana
- **Assets:** native USDT and USDC, already there

4 Two Transaction Types

4.1 Transparent (ordinary Solana)

Speed: Solana's
Cost: fraction of a cent, in SOL
Privacy: none

4.2 Shielded (through Rand)

Speed: Solana's, plus a few seconds to build the proof
Proof generation: 2–4 seconds, on the user's own laptop
Cost: SOL network fee (paid by relayer) + SHRUGG fee
Privacy: sender, recipient, and amount hidden

How it works: a zero-knowledge proof demonstrates that a transfer is valid — the sender owned the funds, nothing was created from nothing — without revealing who, whom, or how much. Solana validators verify the proof without learning its contents.

Note: users generate their own proofs. There is no proving network to pay for, and no GPU required.

5 A Single Token

Rand has one token. Earlier designs had two — ATLAS for staking and gas, SHRUGG for privacy — but once Solana provides consensus there is no validator set for a staking token to secure and no separate gas to price. **SOL fills that role, so ATLAS was removed.**

5.1 SHRUGG — The Confidential Transfer Token

Purpose:

- Pay relayers to submit confidential transfers
- The only way to pay for privacy without revealing who you are

Supply:

- Initial: 200 million tokens
- Emission: 6% declining to 1%, on a fixed published schedule

Why emission is fixed rather than demand-linked: tying issuance to shielded activity would let anyone read private transaction volume off the supply curve. We will not build a side channel into monetary policy.

5.2 Why One Token Is Stronger

1. **No inflation subsidy.** SHRUGG demand comes from users buying it to transact, not from rewards printed to attract validators.
2. **Structural necessity.** Every confidential transfer requires SHRUGG. It is not an optional governance token bolted onto a protocol that works without it.
3. **Nothing to bootstrap.** A staking token needs a validator set, and a validator set needs a token price. We skip that loop entirely.

6 zUSDT: Private Stablecoin, No Peg Risk

6.1 The Terra Lesson

Terra's UST collapsed in May 2022 — \$60B destroyed, LUNA supply inflated 18,500×. The cause was structural: UST's peg was held by a mechanism rather than by reserves, and a mechanism can be attacked.

6.2 zUSDT's Design: No Mechanism At All

zUSDT is fully reserved. Every unit is backed by one unit of native USDT held in a Solana program, redeemable on demand.

- **Backing:** 100% native USDT. Not 90%, not fractional, not partly in our own token.
- **Reserve location:** the same chain as the token — no bridge between the two
- **Verification:** anyone can compare reserves to supply on-chain, continuously
- **Privacy:** transacts confidentially through the Rand pool

What it does not have, deliberately: no collateral ratio to manage, no absorber token to dilute, no redemption rate limits, no monetary policy. A fully-reserved token cannot enter a death spiral, because there is no reflexive loop to spiral.

We abandoned a more ambitious design to get here. An earlier proposal, USDB, was fractionally backed with SOL and SHRUGG absorbing volatility, defended by dynamic collateral ratios and rate-limited redemptions. It was a sound design. But every one of those defences exists to manage the possibility of backing falling short — and full reserving removes that possibility outright. We chose fewer moving parts.

6.3 The Risk We Cannot Remove

Tether can freeze USDT balances, and has done so at law-enforcement request. If our reserve were frozen, redemption would halt. No design that wraps a centrally-administered asset can prevent this, and we would rather state it plainly than bury it. zUSDT offers privacy from observers, not from the issuer. Mitigation is to shard the reserve across vaults so a single freeze impairs only part of redemption capacity.

7 Post-Quantum Security

Encrypted data on a public ledger is permanent. An adversary can record it today and decrypt it whenever quantum computers mature. Rand therefore uses the NIST post-quantum standards where it matters most — for confidentiality.

Component	Typical	Rand	Quantum-safe?
Transfer data encryption	ECDH	ML-KEM (Kyber)	Yes
Spend authorisation	ECDSA	ML-DSA (Dilithium)	Yes
Commitments	Pedersen	Poseidon	Yes
Hashing	SHA-256	SHA-3, BLAKE3	Yes
Proof verification	—	Groth16	No — see below

We are explicit about the exception. Verifying proofs on Solana requires pairing arithmetic that is not quantum-safe. This is a deliberate trade, not an oversight, and the reasoning is that the two risks are not symmetric:

- **Confidentiality cannot be fixed later.** A ciphertext published today is decryptable forever once the maths breaks. So it must be post-quantum now.
- **Verification can be fixed later.** It is enforced by a program we can upgrade. Replace the verifier before quantum computers arrive and nothing historical is at risk.

Spending the post-quantum budget on the irreversible risk is the correct allocation. The residual exposure is that a quantum adversary, acting before we upgrade, could forge a proof and steal from the pool — bounded, detectable through reserve auditing, and mitigated by upgrading. They could not decrypt anyone’s transaction history. Competitors claiming end-to-end post-quantum security on Solana today are either not verifying proofs on-chain or not being precise.

8 Market Opportunity

8.1 Total Addressable Market

- **Cryptocurrency Market:** \$2.5 trillion
- **DeFi:** \$100B+ TVL
- **Privacy coins:** \$5B+ market cap
- **Stablecoins:** \$150B+ market cap
- **Growth:** 40-50% CAGR

8.2 Competitive Landscape

Feature	Zcash	Token-2022	Privacy rollups	Rand
Hides amounts	Yes	Yes	Yes	Yes
Hides participants	Yes	No	Yes	Yes
Native USDT/USDC	No	No	Bridged	Yes
Solana speed	No	Yes	No	Yes
No bridge required	No	Yes	No	Yes
PQ confidentiality	No	No	No	Yes

Our position: the only way to transact native Solana USDT and USDC with full confidentiality, without bridging and without leaving the chain the liquidity is on.

9 Business Model

9.1 Revenue Sources

1. **Confidential transfer fees:** treasury share of every SHRUGG fee
2. **zUSDT volume:** fees on shielded stablecoin transfers
3. **Fee burn:** reduces supply as usage grows

9.2 Value Capture

1. **Usage-driven demand:** every confidential transfer requires buying SHRUGG
2. **Fee burn:** 10% of each fee, tying supply to volume
3. **Treasury:** 20% of fees fund development and audits

No staking yield. There is nothing to stake, because Solana secures the chain. SHRUGG accrues value from demand rather than from emissions — which means slower early distribution, and no inflationary sell pressure.

10 Go-to-Market Strategy

10.1 Phase 1: Developer Community (Months 1-6)

- Open-source with documentation
- \$1-5M grants program
- \$500K-1M hackathons
- Security audits

KPIs: 50+ developers, 10+ apps in development, 3+ audits

10.2 Phase 2: Devnet and Audit (Months 6–12)

- Programs deployed to Solana devnet, then mainnet-beta with deposit caps
- Trusted setup ceremony, run publicly
- \$2M bug bounty; two independent circuit and program audits

KPIs: audits complete, ceremony concluded, 10K+ devnet shielded transfers

10.3 Phase 3: Mainnet with Caps (Month 12)

- Shielded pool live on Solana mainnet, deposits capped and raised on evidence
- zUSDT launched at full reserve
- Relay network open to anyone

KPIs: \$10M shielded TVL, growing anonymity set

Note on caps: a shielded pool is a publicly quantified honeypot, so its balance is the reward for breaking it. Caps that rise with operating history are the responsible way to launch, and we prefer slower growth to an uncapped pool secured only by optimism.

10.4 Phase 4: Growth (Year 2+)

Focus: business payments, payroll, treasury operations, exchange settlement. Extend to native USDC on identical terms.

11 Investment Opportunity

11.1 Funding Requirements

Seed Round: \$5-10M

- Engineering: 50%
- Security: 15%
- Operations: 20%
- Marketing: 15%

Series A: \$20-40M (post-mainnet)

Allocation	Percentage
Team & Advisors	15% (4-year vest)
Foundation/Treasury	20%
Private Sale	10%
Public Sale	10%
Community	10%
Ecosystem	15%
Staking Rewards	20%

11.2 Token Allocation

11.3 Risk Factors & Mitigations

1. **Program defect:** the pool is a concentrated, publicly valued target. Mitigation: deposit caps, two independent audits, formal verification of mint/redeem and state transitions.
2. **Anonymity set size:** privacy is weakest at launch, exactly when users may assume otherwise. Mitigation: communicate this plainly; a single multi-asset pool rather than one per asset.
3. **Solana dependency:** our safety ceiling is Solana's. A consensus failure there could permit a double-spend here. Accepted deliberately in exchange for native assets and no security budget.
4. **Tether freeze:** could halt zUSDT redemption. Mitigation: reserve sharding. Cannot be eliminated.
5. **Quantum forgery before verifier upgrade:** bounded by pool size, detectable via reserve auditing, mitigated by upgrading the verifier.
6. **Regulatory:** privacy tooling attracts scrutiny. Mitigation: no protocol-level anonymity for the issuer, optional viewing keys for holders needing to disclose to auditors or regulators, KYC at fiat on/off-ramps.

12 Key Success Metrics

Metric	Year 1	Year 2	Year 3
Shielded TVL	\$10M+	\$150M+	\$1B+
Confidential transfers/day	1K+	25K+	250K+
Anonymity set (active notes)	50K+	1M+	10M+
zUSDT supply	\$5M+	\$75M+	\$500M+
Active relayers	10+	50+	200+
SHRUGG market cap	\$50M+	\$250M+	\$750M+

The metric that matters most is **anonymity set size**. TVL can be supplied by a handful of large depositors while leaving privacy weak; a large set of active notes cannot. We report both.

13 Why Rand Will Win

1. **We start where the liquidity is.** Competitors must attract stablecoin issuers, validators, and liquidity to a new chain before being useful. Native USDT and USDC are already on Solana.
2. **No bridge.** The largest losses in crypto are bridge failures. Reserve and token share a chain, so there is no bridge to attack.
3. **Structural token demand.** Every confidential transfer must buy SHRUGG to pay a relay. Demand tracks usage, not emissions.
4. **Honest post-quantum positioning.** We are post-quantum where it is irreversible and explicit about where we are not. That distinction is defensible under scrutiny; blanket claims are not.
5. **Small surface.** Three programs and a fee token. No consensus protocol, no validator set, no staking, no stablecoin peg mechanism — each an entire class of failure we do not carry.

14 Contact

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*This document provides a high-level overview for investment consideration.
Detailed technical specifications and due diligence materials available upon request.*